INCH-POUND

MIL-DTL-17361/1B(SH)
19 September 2006

SUPERSEDING
MIL-C-17361/1A(SH)
13 March 1995

DETAIL SPECIFICATION SHEET

CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPES AQB-A50, AQB-A51, NQB-A50, AND NQB-A51

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17361.

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TABLE I. Requirements.

Maximum rated continuous current	50 amperes			
Voltage and frequency		Applicability		
		A50	A50D	A51
	500 VAC, 60 Hz	X		X
	500 VAC, 400 Hz	X		
	250 VDC	X	X	
Interrupting rating		Applicability		
		A50	A50D	A51
	12,000 amperes, DC		X	
	5,000 amperes, 60 Hz	X		X
	2,500 amperes, 400 Hz	X		
	2,500 amperes, DC	X		
Peak let-through current	Not applicable			
Trip unit	Non-interchangeable			
Current limiting fuses	Not applicable			
Type of connections	Back (removable mtg.)			
Electric operator	Not available			
Undervoltage trip or shunt trip	Neither available			
Auxiliary switch	Not available			
Handle lock	Group 1 (Figure 1 of MIL-DTL-17361)			
Additional identification	Front cover – “On-Off” Handle – ampere rating			
Additional mounting requirements	Each circuit breaker shall have two threaded metal inserts molded in the cover for insertion of the supporting screws. Stud terminals or lug terminals connected to the circuit breaker shall not turn to reduce the electrical clearances. Stud terminals or lug terminals shall be supplied with mounting bases.			
Shock test	Mounting shall be in a vertical position in accordance with fixture 6E of MIL-S-901			
Handle boot	See Figure 3 and Table V			

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TABLE II. Instantaneous trip calibrations.

Type designation	Trip element rating	Test current amperes	Trip time /1 (seconds)
AQB-A50 60 Hz, 400 Hz	10, 15, 20	30X (element rating)	0.01–0.70
	25, 30	600	0.01–1.30
	35, 40, 50	600	0.01–3.00
AQB-A50 DC	10, 15, 20	40X (element rating)	0.01–0.40
	25, 30	600	0.01–1.10
	35, 40, 50	600	0.01–2.50

NOTE:

/1 Test each pole individually; test current shall be preset and maintained within ±10% of specified value until circuit breaker trips.

TABLE III. Instantaneous trip calibration (for AQB-A51 only).

Type designation	Trip element rating	Test current amperes	Trip condition /1
AQB-A51	All ratings	20X	No Trip
		40X	Trip

NOTE:

/1 Test each pole individually, test current shall be a current pulse of 80–100 milliseconds duration having no more than 5 percent current asymmetry in the first half cycle.

TABLE IV. Tripping characteristics – long-time delay (LTD).

Type designation	Current /1 % element rating	Tripping time	
		Minimum	Maximum
AQB-A50 (all elements)	115	60 minutes	No limit
	138	200 seconds	60 minutes
	200	10 seconds	200 seconds
AQB-A51 (all elements)	115	60 minutes	No limit
	138	200 seconds	60 minutes
	200	10 seconds	200 seconds
	400	4 seconds	16 seconds
	800	0.8 seconds	4 seconds

NOTE:

/1 Test current shall be through all phases in series.

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TABLE V. Rubber handle boot.

Properties	
Color	Clear transparent
Specific gravity at 77 °F	1.15±0.02
Chemical classification	A-A-59588
Shore A hardness	50±5
Minimum tensile strength (lb/in^2)	1100
Minimum elongation (%)	450
Minimum tear strength (lb/in^2)	150
Minimum bayshore resiliency (%)	40
Fatigue life at 125% strength (flexes)	800,000
Test requirements	
Color	/1
Shore A hardness	ASTM D2240
Tensile strength	ASTM D412
Elongation	ASTM D412
Tear strength	ASTM D624
Bayshore resiliency	ASTM D412
Fatigue life	ASTM D412

NOTE:

/1 A 0.075-inch slab of the rubber shall be placed in direct contact with a piece of newsprint. The rubber shall not interfere with the ability to read the newsprint.

TABLE VI. Test requirements.

Electrical endurance test	The rate of operation shall be 6 operations per minute for a total of 6000 close-open operations.			
Temperature rise tests	Size of cable and bus bars for temperature rise tests			
	Copper rating (amperes)	Cable size		Copper bus size (inch)
		AC	DC	AC and DC
	50	AWG 7	AWG 9	1/2 X 1/8

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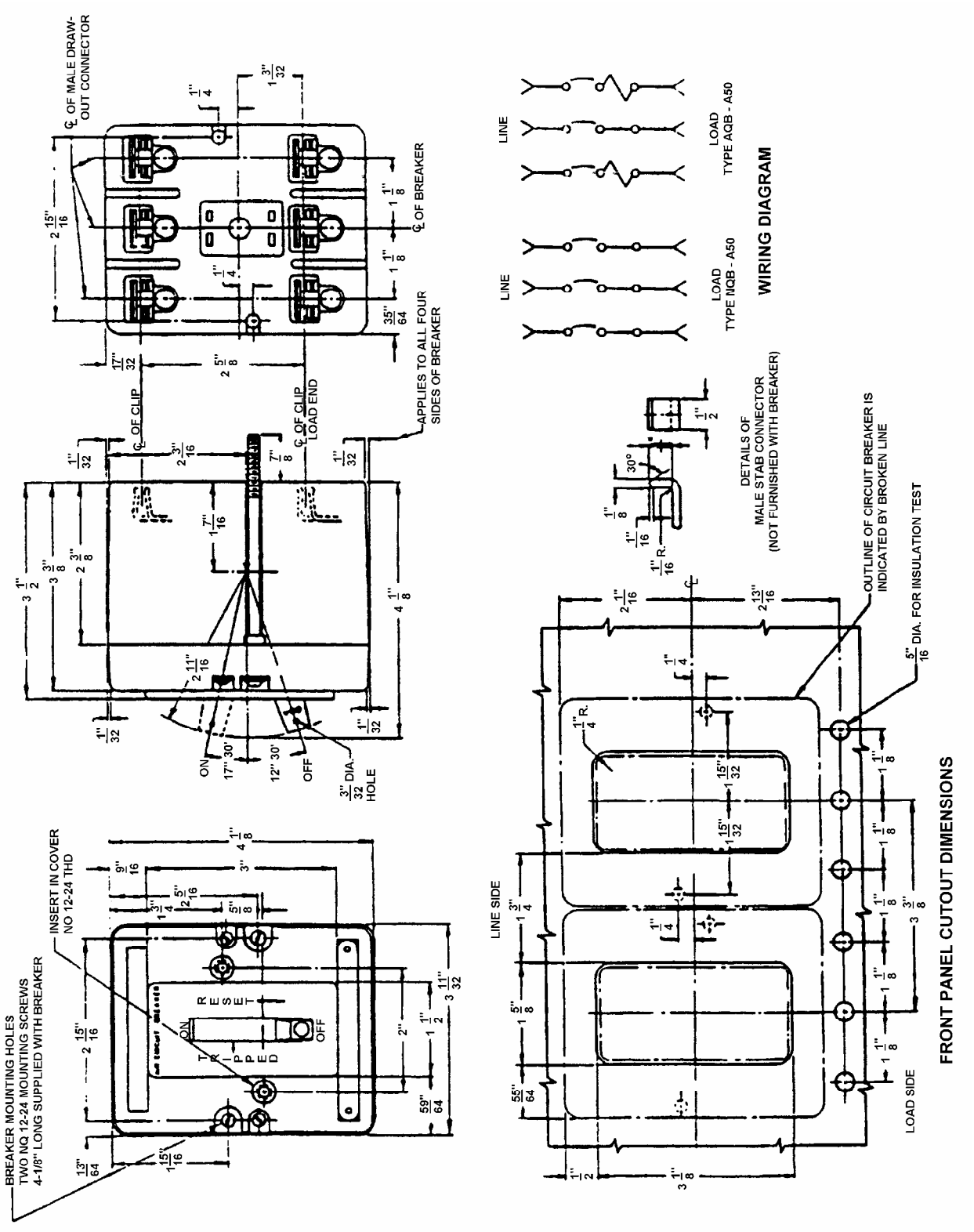


FIGURE 1. Circuit breaker types AQB-A50, AQB-A51, NQB-A50, and NQB-A51 outline and mounting dimensions.

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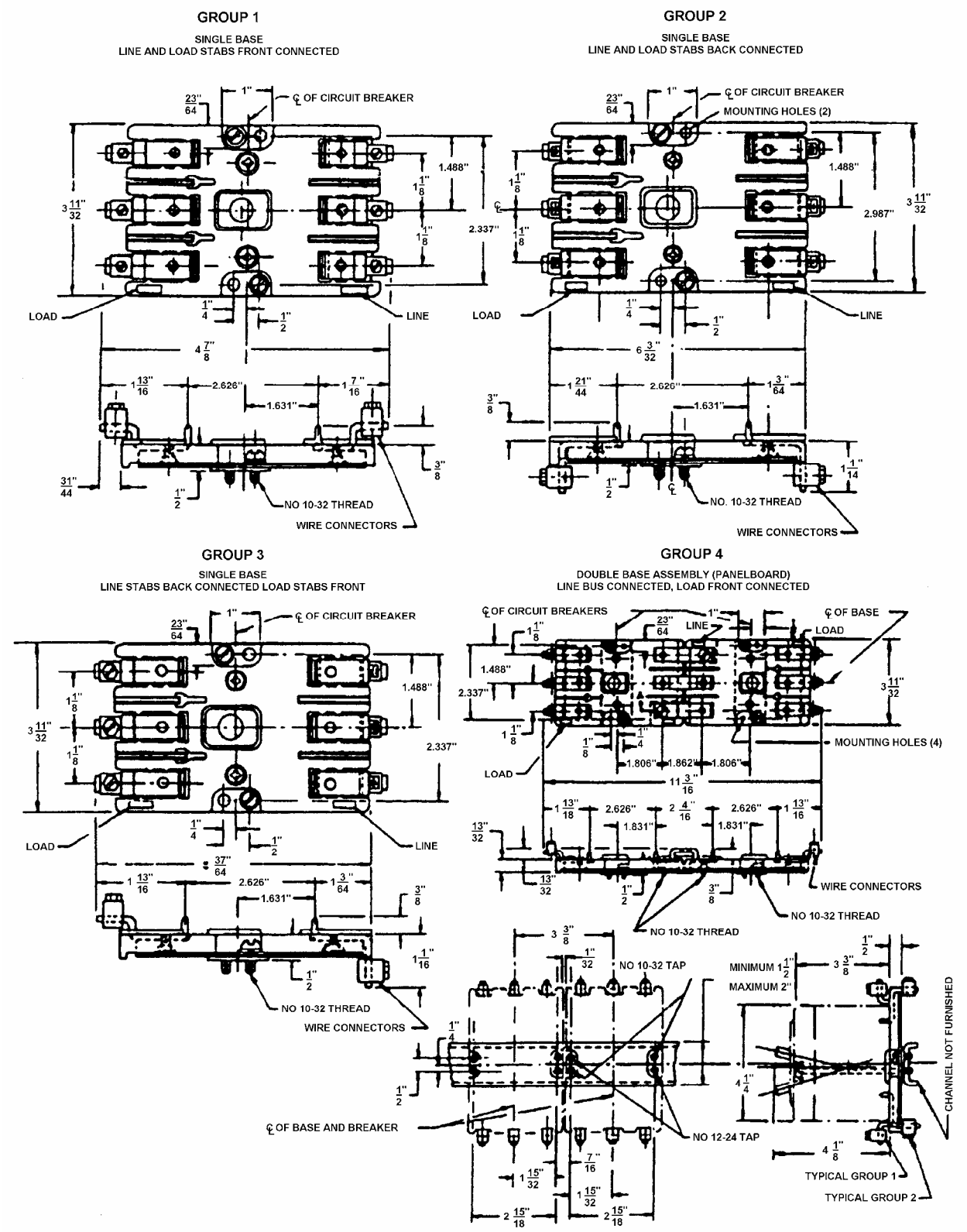


FIGURE 2. Mounting bases for circuit breaker types AQB-A50, AQB-A51, NQB-A50, and NQB-A51 outline and mounting dimensions.

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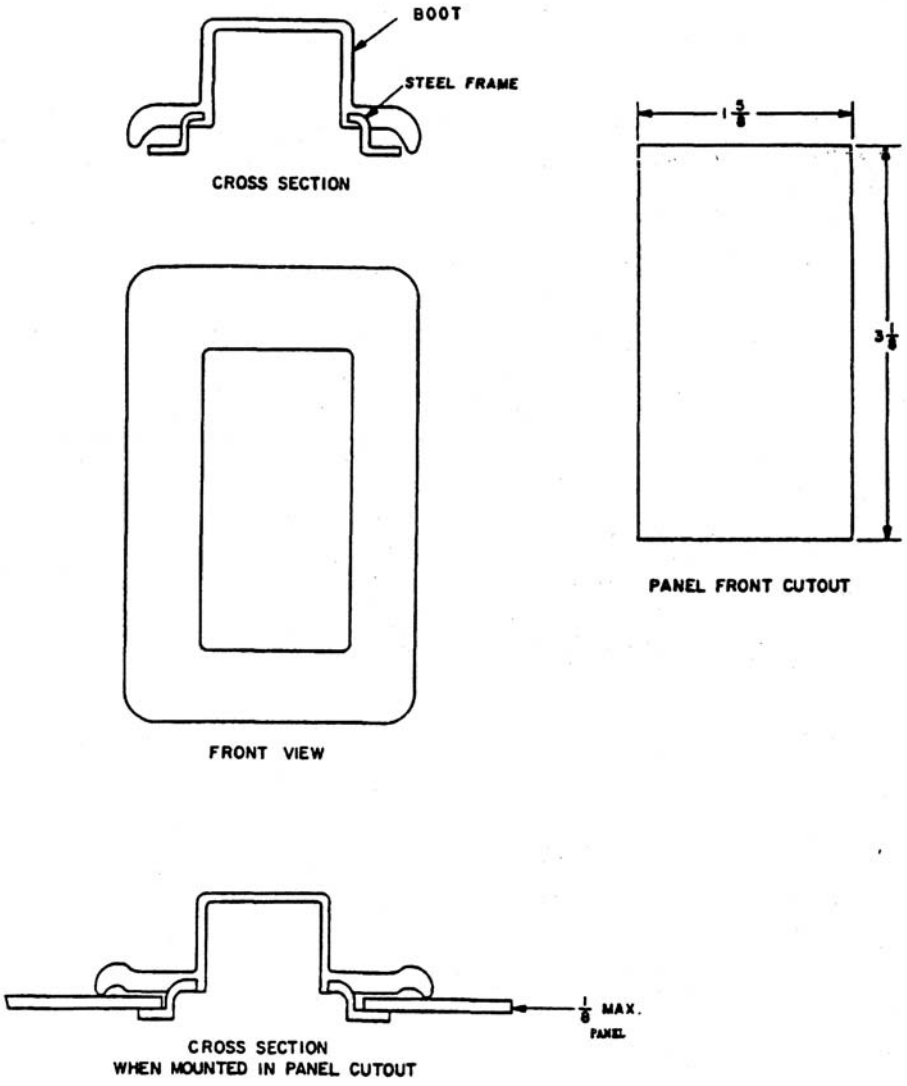


FIGURE 3. Circuit breaker handle boot for AQB-A50, AQB-A51, NQB-A50, and NQB-A51.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy – SH
(Project 5925-N185-001)

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